LESSON 8.6

Solve and Graph Inequalities

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve an inequality and graph its solution set on a number line.

WHY THIS MATTERS Detective Kim knows that the suspect withdrew more than $50 from an ATM. The amount plus a $3 fee totaled more than $53. She writes x + 3 > 53 and needs to solve for x, then graph the solution set to share with the bank. How can she find and display all possible withdrawal amounts?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Solve *Spanish: Resolver*	To find the number that makes it true.	x + 3 > 10 → subtract 3 → x > 7 — any number greater than 7 works
Graph *Spanish: Graficar*	To show answers on a number line with circles and shading.	x > 7: open circle at 7, shade right — shows all solutions at a glance
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	x = 8 is a solution to x > 7 because 8 > 7 is true; x = 5 is not because 5 > 7 is false
Substitute *Spanish: Sustituir*	To put a number in place of a letter.	Substitute x = 5 into x + 3 > 10: 5 + 3 = 8, and 8 > 10 is false, so x = 5 is not a solution
Solution set *Spanish: Conjunto solución*	All the numbers that make the inequality true.	For x > 4, the solution set is every number greater than 4, such as 5, 6, 7, ...
Inverse operation *Spanish: Operación inversa*	Two math actions that undo each other, like × and ÷.	To solve x + 3 > 7, use the inverse of adding 3: subtract 3 to get x > 4.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Solve x + 3 > 53. The inverse of adding 3 is subtracting 3.

1. I subtract 3 from both sides: x + 3 − 3 > 53 − 3, so x > 50.
2. The symbol > does not include 50, so I use an open circle at 50.
3. 'Greater than' shades right. Check: x = 60 gives 60 + 3 = 63 > 53. True!

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Solve $x - 5 \leq 8$. What is the inverse of subtracting 5?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Solving an inequality is like solving an equation because I still use _____, but it is different because the answer is _____ instead of _____.

Word bank Solve | Graph | Solution | Substitute | Solution set | Inverse operation | inverse operations | many values | one value | graph | solution set | number line

WATCH OUT!

A common mistake

A common mistake in Solve and Graph Inequalities is shading the number line in the wrong direction after solving. For example, a student solves $x + 4 > 10$ correctly to get $x > 6$, but then shades LEFT toward smaller numbers instead of RIGHT toward greater numbers. The shading direction always matches the solved symbol, not the original problem: '>' and '≥' shade right (greater numbers), '<' and '≤' shade left (smaller numbers)

LESSON 8.6

Solve and Graph Inequalities

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Solve x + 3 > 9, then graph the solution on the number line.



HINT Subtract 3 from both sides: x > 6.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Solve: x + 6 > 14

(A) x > 8
(B) x < 8
(C) x = 8
(D) x > 20

HINT Solve it like an equation: x has 6 added to it. The > symbol just rides along.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 9 \leq 3$

(A) $x \leq 6$

(B) $x \leq 12$

(C) $x \geq 12$

(D) $x = 12$

HINT x has 9 subtracted from it. Undo that first; the ≤ symbol keeps its direction.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: $x + 10 \geq 25$

(A) $x \geq 15$

(B) $x \leq 15$

(C) $x = 15$

(D) $x \geq 35$

HINT What's being done to x? 10 is added. Plan the inverse operation before you look at the choices.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 4 < 11$

(A) $x < 7$

(B) $x < 15$

(C) $x > 15$

(D) $x = 15$

HINT x has 4 taken away. Which operation puts that 4 back?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Catch the Wrong Inverse Operation

1. **Inequality** $x + 6 > 14$
2. **Inverse operation** Add 6 to both sides
3. **Simplify** $x > 20$
4. **Graph** Open circle at 20, shade right

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT The inequality has + 6. To undo adding 6, do you add or subtract?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

SENTENCE STARTER Solving an inequality is like solving an equation because I still use ___, but it is different because the answer is ___ instead of ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.6

Solve and Graph Inequalities

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each inequality, describe the inverse operation, and describe the graph.

Inequality	Inverse Operation	Solution	Graph Description
x + 11 ≥ 20			
x − 3 < 7			
x + 8 ≤ 20			

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Solve each inequality, then test if x = 10 is a solution.

Groups: **x = 10 IS a solution** **x = 10 is NOT a solution**

_____ x + 2 > 8

_____ x − 5 ≥ 3

_____ x + 1 ≤ 15

_____ x + 3 > 15

_____ x − 2 ≥ 9

_____ x + 6 < 10

3 **Decide whether each pair balances. Check Yes or No, then explain.**

Solve the inequality x + 7 ≥ 19 using the balance scale. What is the boundary value?

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Is x = 5 a solution to x + 4 > 12?

(A) No, because 5 + 4 = 9, and 9 is not greater than 12

(B) Yes, because 5 + 4 = 9

(C) Yes, because 5 > 4

(D) No, because 5 < 12

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Catch the Open vs. Closed Circle Mistake

1. **Inequality** x − 4 ≥ 9
2. **Inverse operation** Add 4 to both sides
3. **Simplify** x ≥ 13
4. **Graph** Open circle at 13, shade right

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

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LESSON 8.6

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Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each inequality and verify by testing two values.

Inequality	Solution	Test x = 5	Test x = 20
x + 8 > 15			
x − 6 ≤ 10			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Solving Error

1. **Inequality** $x - 8 \geq 15$
2. **Inverse operation** Subtract 8 from both sides
3. **Simplify** $x \geq 7$
4. **Graph** Closed circle at 7, shade right

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A detective has a budget of at most $100 for supplies. She already spent $37 on gloves. Write an inequality for the remaining amount r she can spend, solve it, graph it, and name three possible values for r.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

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